

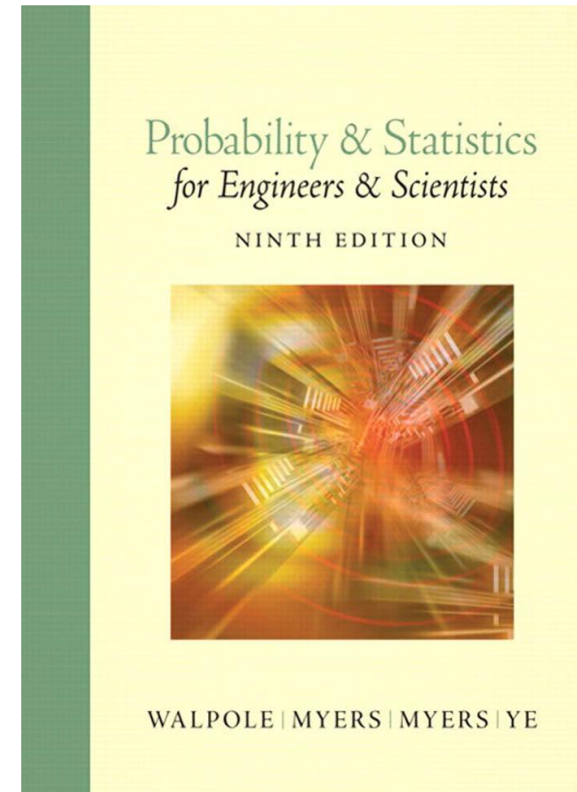
MTH-262 Statistics and Probability Theory

Lecture 01



MTH-262 Statistics and Probability Theory

- Credit hours 3(3-0)
- Quizzes 04 (15 Marks)
- Assignments 04 (10 Marks)
- Mid Marks 25 Marks
- Final 50 Marks



- Recommended Book: “Probability and Statistics for Engineers and Scientists” by “Ronald E. Walpole” 9th Ed.

Course Outline

- Introduction to statistics and statistical methods, Frequency Distributions & Representation of data, Measure of Central tendency, Measures of Dispersion, Probability theory, Counting Rules, Conditional Probability, Law of total probability and Bays Rule, Concept of Discrete and Continuous Random variable, Cumulative distributions, Joint probability Distributions, Uniform, Binomial and, Poisson, & Geometric Distributions, Uniform & Normal Distribution, Gamma, Exponential distributions, Simple linear Regression and fitting of Curves. Correlation study. Testing about population Mean, proportion for one sample and two samples. Confidence interval for population Mean, proportion for one sample and two samples.

Statistics (First Part of this Course)

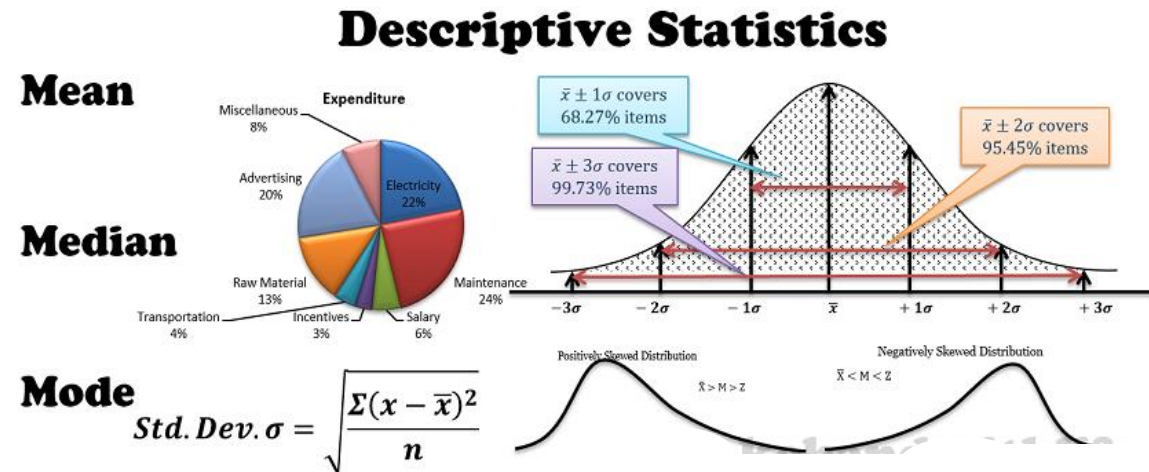
- Collection (Primary, Secondary) → Simple Random Sampling (Article 1.2)
- Presentation (Graphs and Charts)
- Analysis (Statistical tools are used to analyze data)
- Interpretation (A statement about the analyzed results)

Definition: Statistics is a branch of applied mathematics that involves the collection, description, analysis, and inference of conclusions from quantitative data. The mathematical theories behind statistics rely heavily on differential and integral calculus, linear algebra, and probability theory.

Statistics (Branches of Statistics)

Descriptive Statistics

Descriptive statistics consists of methods for organizing and summarizing information. Descriptive statistics includes the construction of graphs, charts, and tables and the calculation of various descriptive measures such as averages, measures of variation, and percentiles.



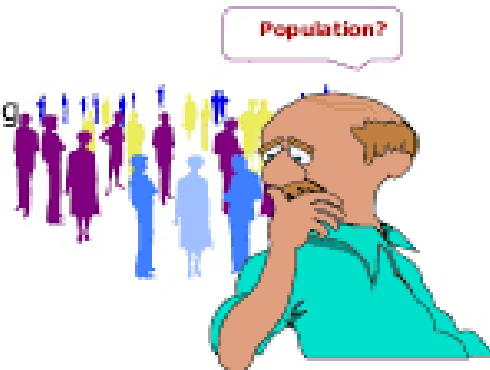
Statistics (Branches of Statistics)

Inferential Statistics

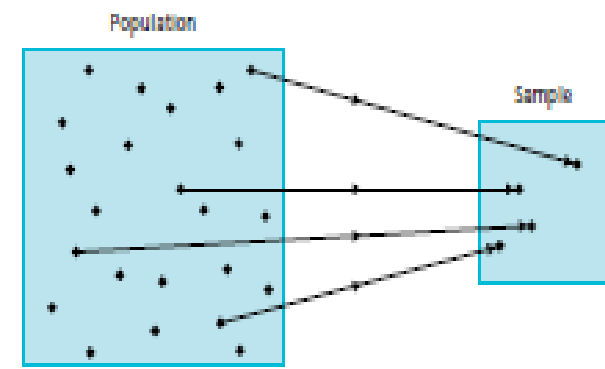
Inferential Statistics consists of methods for drawing and measuring the reliability of conclusions about a population based on information obtained from sample of the population.

INFERENCE STATISTICS

1. Involves
 - Estimation
 - Hypothesis Testing
2. Purpose
 - Make Decisions About Population Characteristics



Statistics (Some Definitions)



Population

Collection of all objects relating with the subject of study is known as population.

Sample

A representative part of the population.

Parameter

Any measure taken from the population data is called “Parameter”.

Statistic

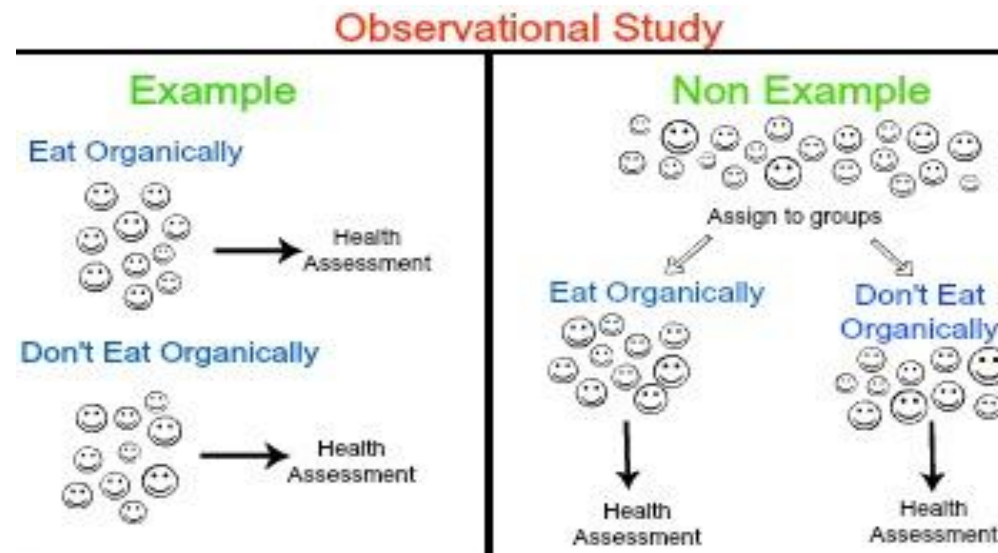
Any measure taken from the sample data is called “Statistic”.

Statistics

We often need to classify statistical studies as either observational studies or designed experiments.

Observational Studies

In an observational study, researchers simply observe characteristics and take measurements, as in a sample survey.



Observational Study

Left side (Example: Observational Study)

- People are already in two groups:
 - Those who eat organically
 - Those who don't eat organically
- The researcher does not assign anyone to a group.
- They simply observe what people normally do and then perform a health assessment on each group.

Key idea:

You observe existing behavior and compare outcomes.

Because there is no random assignment, other factors (exercise, income, lifestyle, etc.) might affect health. So you can find association, but not strong cause-and-effect.

Designed Experiment

Right side (Non-Example here = An experiment)

- First, participants are assigned to groups:
 - One group is told to eat organically
 - The other group is told to not eat organically
- Then both groups get a health assessment.

Key idea:

The researcher controls who goes into which group (usually randomly).

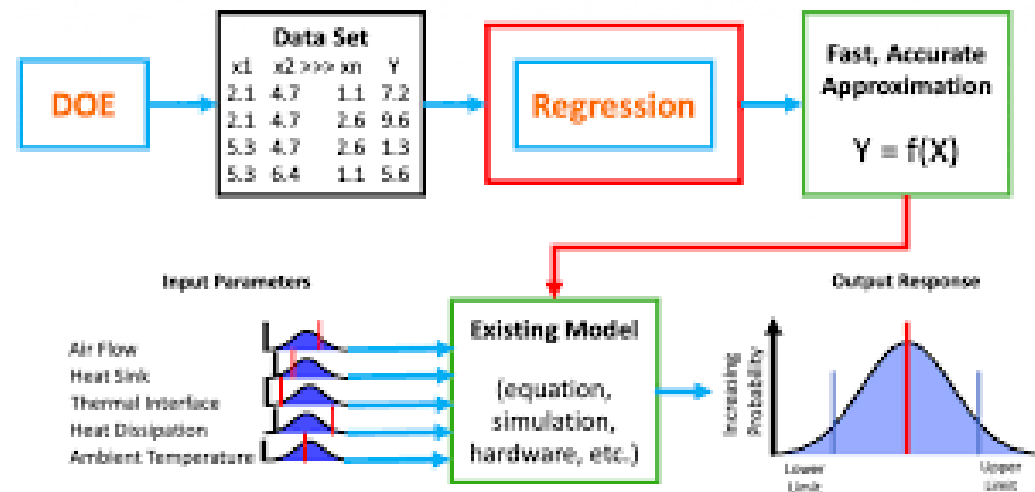
This is an experimental study, not observational. Because of assignment, it's much easier to make causal conclusions (does organic food *cause* better health?).

Statistics

We often need to classify statistical studies as either observational studies or designed experiments.

Designed Experiments

In a designed experiment, researchers impose treatments and controls and then observe characteristics and take measurements. Designed experiments can help establish causation



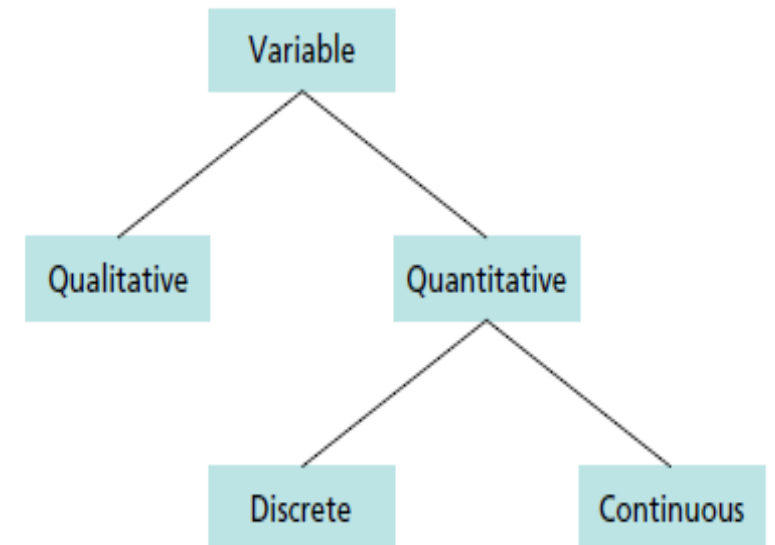
Statistics —————> Data

Data

The collection of all observations for a particular variable is called a data set.

Variable

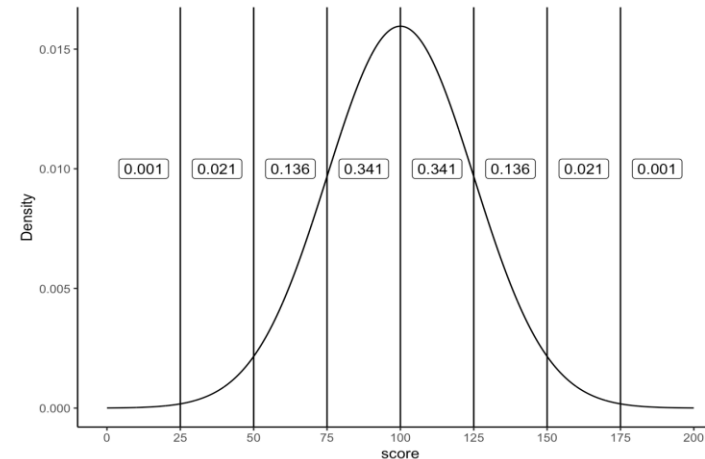
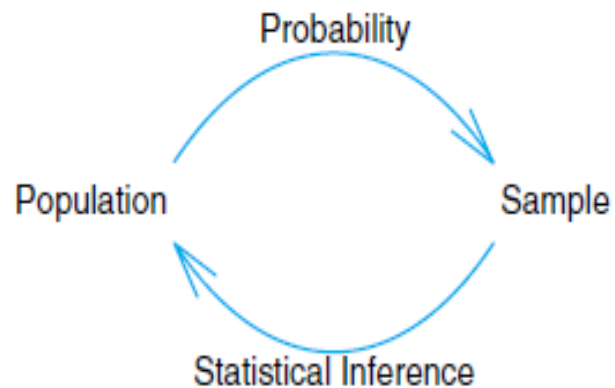
A characteristic that varies from one person or thing to another.



Probability Vs. Statistical Inference

Statistical inference makes use of concepts in probability

- The sample along with inferential statistics allows us to draw conclusions about the population, with inferential statistics making clear use of elements of probability.
- Elements in probability allow us to draw conclusions about characteristics of hypothetical data taken from the population, based on known features of the population.



Data Collection

There are two methods for the data collection:

- Primary Collection

Surveys (Telephone surveys, Questionnaire, Through email, Web surveys),
Personal Interviews.

- Secondary Collection

The data available on any authentic source is used for desired purpose

1. <https://stackoverflow.com/>
2. <https://www.kaggle.com/>, etc.
3. <https://www.pbs.gov.pk/>

Data Presentation

Statistical Modelling, Scientific Inspection, and Graphical Diagnostics (Section 1.6)

- Histogram (Presentation of the data and Test for normality)
- Box-and-Whisker Plot/Box-Plot (Detects the outlier)
- Stem-and-Leaf Plot (Present the data and do lose information about any individual)
- Scatter Plot (Functional relationship of variables)

Graphical Diagnostics

Histogram

Histogram helps us to identify the data behavior.

How to Construct a Frequency Distribution

- 1: Sort the data in ascending order.
- 2: Calculate the range of data.
- 3: Decide on the number of intervals in the **frequency distribution**.
- 4: Determine the intervals.
- 5: Tally and count the observations under each interval.

Table 1.4: Car Battery Life

2.2	4.1	3.5	4.5	3.2	3.7	3.0	2.6
3.4	1.6	3.1	3.3	3.8	3.1	4.7	3.7
2.5	4.3	3.4	3.6	2.9	3.3	3.9	3.1
3.3	3.1	3.7	4.4	3.2	4.1	1.9	3.4
4.7	3.8	3.2	2.6	3.9	3.0	4.2	3.5

Table 1.7: Relative Frequency Distribution of Battery Life

Class Interval	Class Midpoint	Frequency, f	Relative Frequency
1.5–1.9	1.7	2	0.050
2.0–2.4	2.2	1	0.025
2.5–2.9	2.7	4	0.100
3.0–3.4	3.2	15	0.375
3.5–3.9	3.7	10	0.250
4.0–4.4	4.2	5	0.125
4.5–4.9	4.7	3	0.075

Graphical Diagnostics

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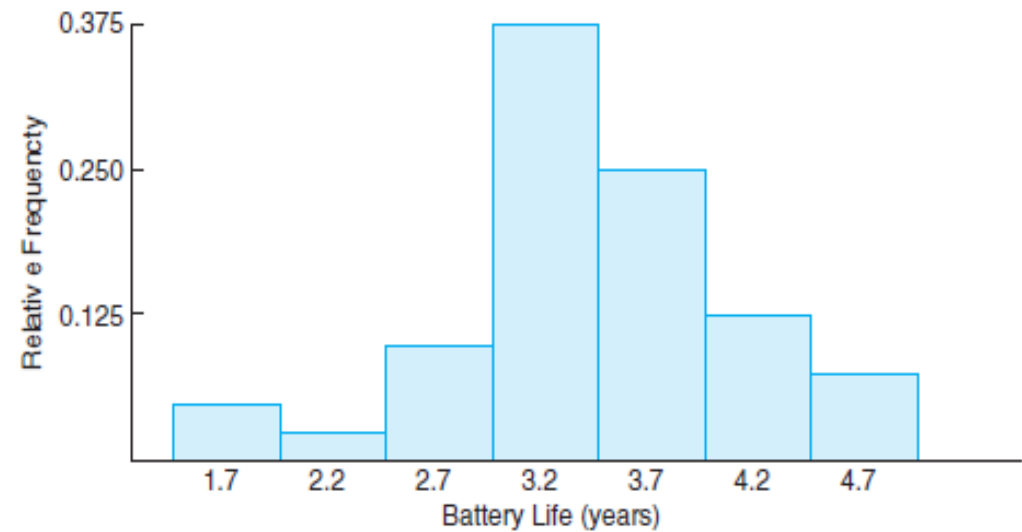


Figure 1.6: Relative frequency histogram.

Graphical Diagnostics

Skewness

Skewness is the departure from symmetry.

(a) Positively skewed **(b)** Normally distributed **(c)** Negatively Skewed

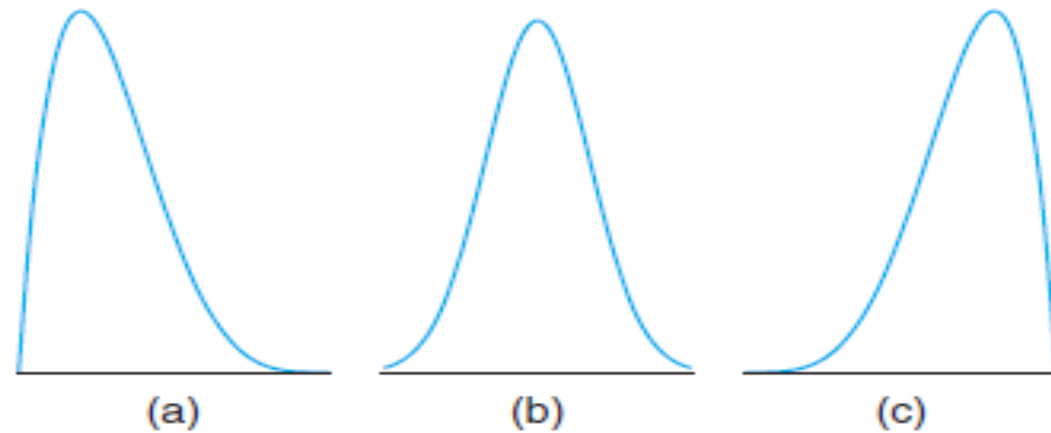


Figure 1.8: Skewness of data.

Graphical Diagnostics

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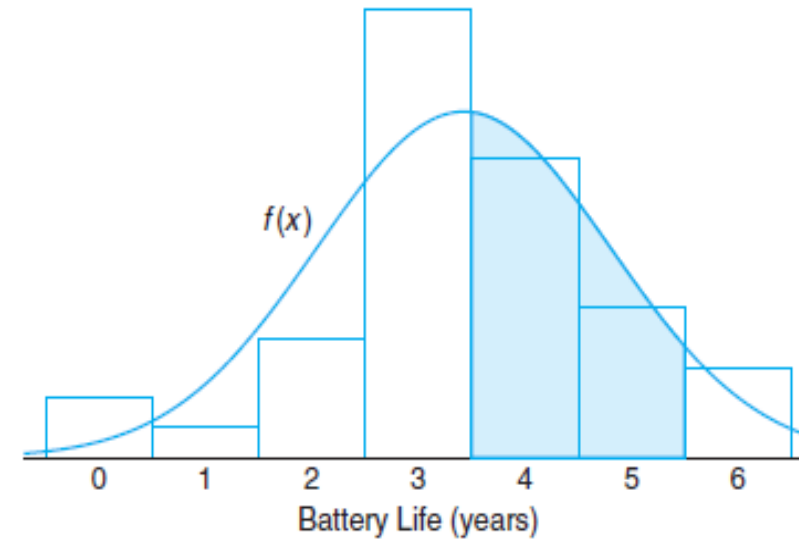
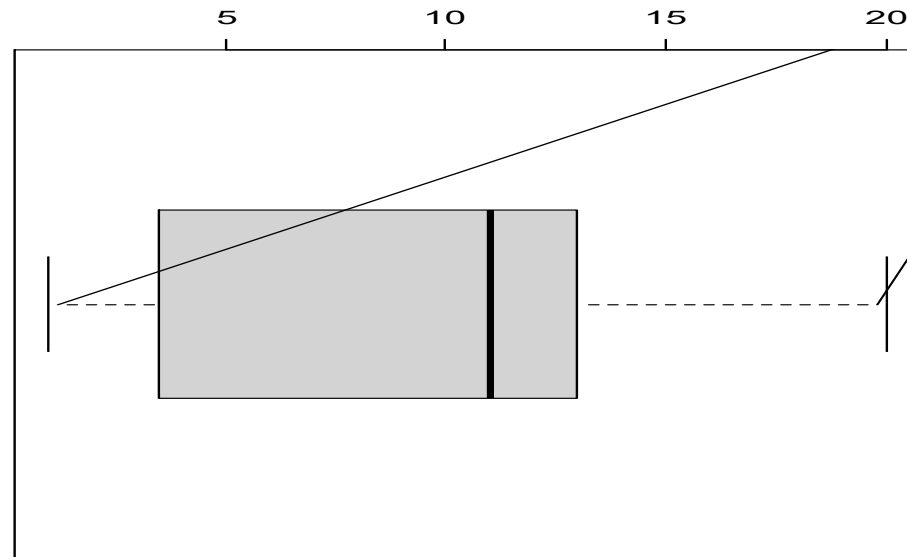


Figure 1.7: Estimating frequency distribution.

Graphical Diagnostics

Box-and-Whisker Plot

1. Five number summary (Lowest value of the data, Largest value of the data, First and Third Quartile, Median)



Graphical Diagnostics

Box-and-Whisker Plot

2. Mainly used to detect the outliers. (If the distance from the box exceeds 1.5 times the interquartile range (in either direction), the observation may be labeled an outlier).

3. Degree of Asymmetry.

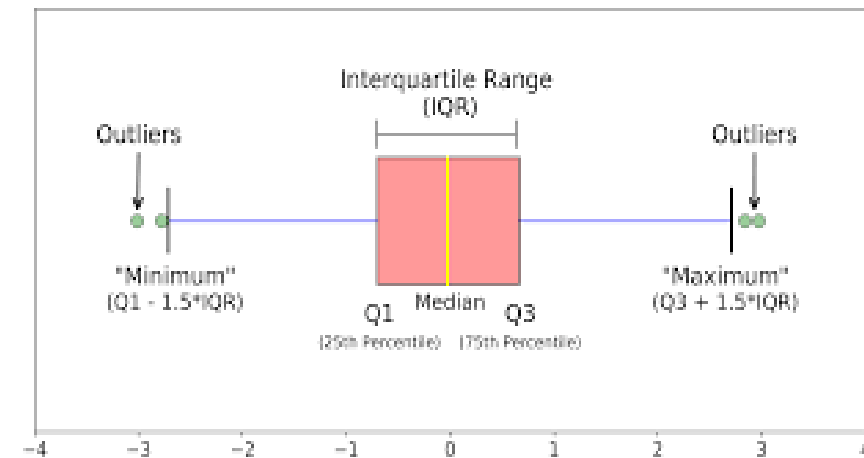


Table 1.8: Nicotine Data for Example 1.5

1.09	1.92	2.31	1.79	2.28	1.74	1.47	1.97
0.85	1.24	1.58	2.03	1.70	2.17	2.55	2.11
1.86	1.90	1.68	1.51	1.64	0.72	1.69	1.85
1.82	1.79	2.46	1.88	2.08	1.67	1.37	1.93
1.40	1.64	2.09	1.75	1.63	2.37	1.75	1.69

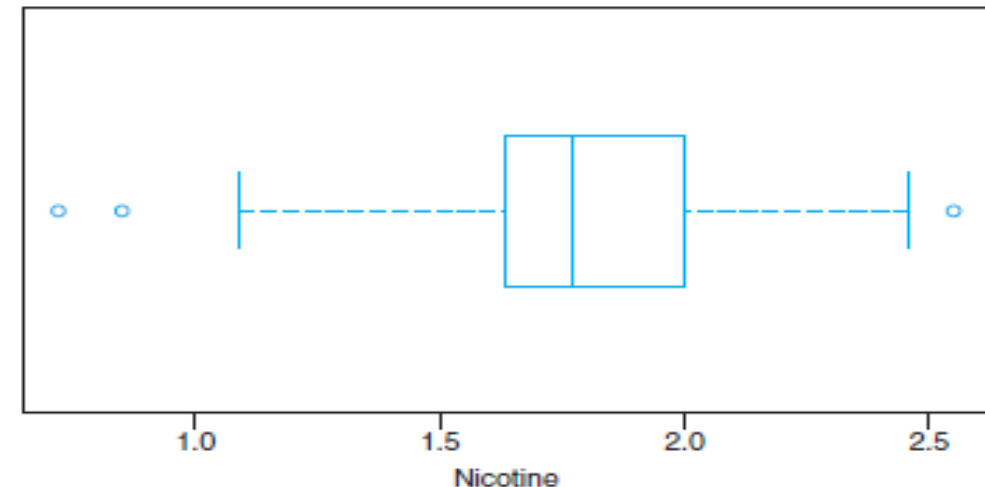
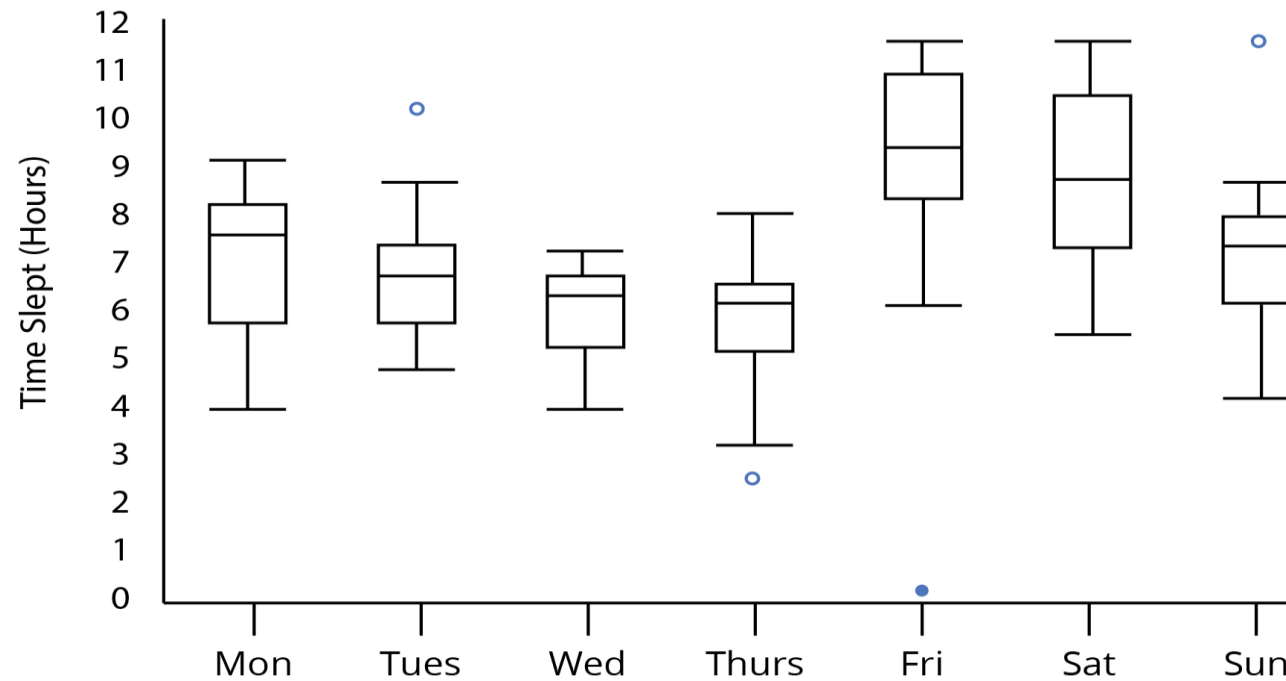


Figure 1.9: Box-and-whisker plot for Example 1.5.

Graphical Diagnostics

Box-and-Whisker Plot

4. More than one series can be compared by using Box-and-Whisker plot



Graphical Diagnostics

Stem-and-Leaf Plot

1. Combined display (Tabular and Graphic)
2. It shows the frequency of an interval and observations belonging to the interval (It doesn't lose the information about the individual observation)

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3.3	3.1	3.7	4.4	3.2	4.1	1.9	3.4
4.7	3.8	3.2	2.6	3.9	3.0	4.2	3.5

Table 1.5: Stem-and-Leaf Plot of Battery Life

Stem	Leaf	Frequency
1	69	2
2	25669	5
3	0011112223334445567778899	25
4	11234577	8

Graphical Diagnostics

Scatter Plot

1. Model Selection
2. Causation

Table 11.1: Measures of Reduction in Solids and Oxygen Demand

Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)	Solids Reduction, x (%)	Oxygen Demand Reduction, y (%)
3	5	36	34
7	11	37	36
11	21	38	38
15	16	39	37
18	16	39	36
27	28	39	45
29	27	40	39
30	25	41	41
30	35	42	40
31	30	42	44
31	40	43	37
32	32	44	44
33	34	45	46
33	32	46	46
34	34	47	49
36	37	50	51
36	38		

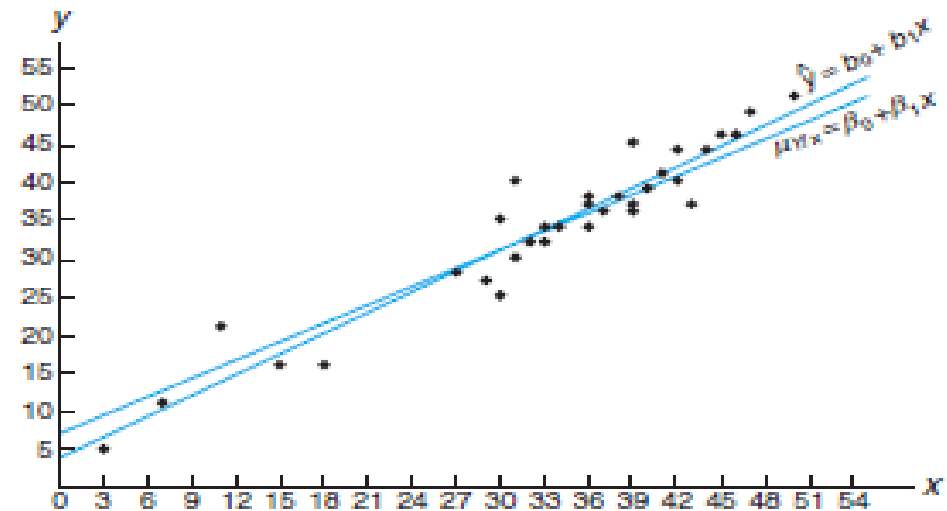


Figure 11.3: Scatter diagram with regression lines.

Graphical Diagnostics

Scatter Plot

1. Model Selection
2. Causation

Table 1.3: Tensile Strength

Cotton Percentage	Tensile Strength
15	7, 7, 9, 8, 10
20	19, 20, 21, 20, 22
25	21, 21, 17, 19, 20
30	8, 7, 8, 9, 10

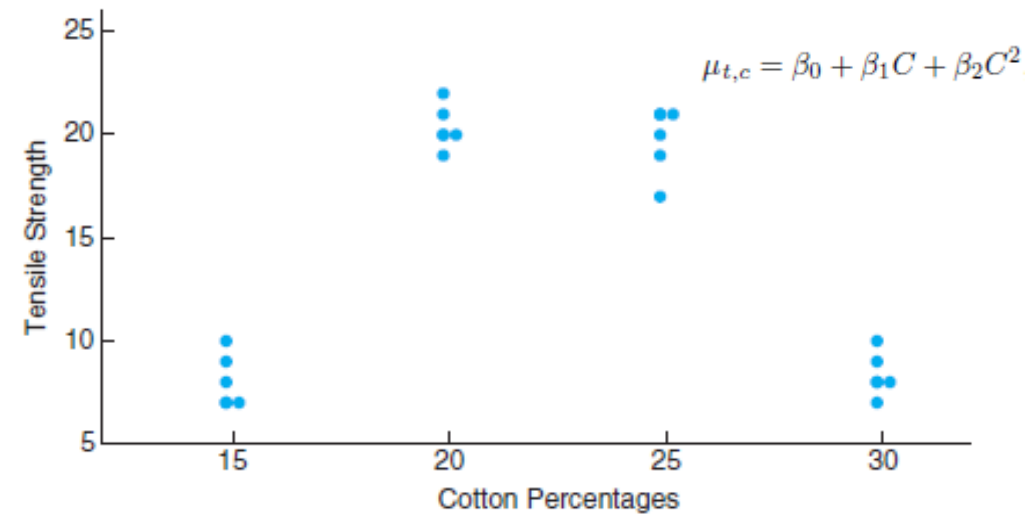


Figure 1.5: Scatter plot of tensile strength and cotton percentages.